

GHUFRAN SHAKEEL

Network Solution Architect | Network Engineer | ISP, Enterprise, Data Center Networks

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Summary

Network Solution Architect specializing in enterprise, service-provider and data-center infrastructure, with hands-on expertise across IP/MPLS, BGP, MPLS L3VPN, EVPN-VXLAN, MP-BGP, SD-WAN and network security Experienced in engineering scalable, resilient and highly available networks spanning underlay and routing foundations, overlay fabrics, secure connectivity and operational platforms.

Core Technical Expertise

Network Architecture & Data Center: Enterprise & Service Provider Networks • ISP Architecture • Data Center Architecture • Leaf-Spine/Clos • Underlay/Overlay • VXLAN • EVPN • MP-BGP • VTEP • VNI • L2VNI • L3VNI • Anycast Gateway • IRB • EVPN Multihoming/ESI • DCI • Network Segmentation • HA/Redundancy • HLD/LLD

Routing, Switching & MPLS: BGP • MP-BGP • IS-IS • OSPF • EIGRP • Route Reflectors • BFD • ECMP • Traffic Engineering • MPLS • L3VPN • L2VPN • VRF • PE-CE • LISP • VLAN • STP/RSTP/MSTP • EtherChannel

WAN, Security & Connectivity: SD-WAN • DMVPN • IPsec • GRE • FlexVPN • WireGuard • SSL VPN • Multi-Site Connectivity • Palo Alto • Fortinet • Cisco ASA • F5 • NGFW • ACL • IPS/IDS • NAT/PAT • AAA • TACACS+ • PKI

Service Provider, Telecom & 5G: IP/MPLS • GPON • CGNAT • ZTP • TR-069/CWMP • GenieACS • CPE Lifecycle Management • ISP Infrastructure • 5G • CBRS • Open5GS • free5GC • UERANSIM

Infrastructure, Virtualization & Automation: VMware ESXi • Proxmox • Docker • Kubernetes • Linux • NIC Teaming • Python • Bash • Ansible • Terraform • Git • YAML • IaC • Network Automation

Observability, Operations & Communications: SolarWinds • Grafana • Zabbix • Prometheus • PRTG • Cacti • SNMP • Syslog • JumpServer • Wireshark • TCPDump • Packet Analysis • RCA • SLA Operations • Network Optimization • SIP • RTP • SBC • QoS • 3CX • Grandstream UCM • ViciDialer

Experience

Multinet

March 2025 – Present

Network Engineer

- Operate and support nationwide IP/MPLS backbone infrastructure delivering secure L2/L3 VPN services to enterprise and banking customers.
- Engineer BGP, OSPF, IS-IS, MPLS L3VPN, VRF and PE-CE routing, resolving complex L2/L3 service-provider and enterprise network issues.
- Drive critical incident resolution, fault isolation and RCA within SLA, maintaining network availability and service continuity.
- Execute network migrations, upgrades and maintenance in coordination with Core Engineering, NOC, Security, Field Operations and customers.
- Monitor network performance and availability through SolarWinds, Grafana, Zabbix and JumpServer, identifying and addressing infrastructure issues.
- Support GPON and enterprise last-mile services while contributing to capacity planning, redundancy and infrastructure optimization.
- Develop technical procedures, implementation documentation and RCA reports supporting operational governance and continuous improvement.

Imperial Technologies Inc

Aug 2025 – Present (freelance)

Network Implementation Engineer

- Architected ISP, enterprise and data-center network environments, designing scalable, secure and highly available infrastructure across service-provider and business networks.
- Designed and validated Leaf-Spine data-center fabrics using underlay/overlay architectures, VXLAN, EVPN, MP-BGP, VTEPs, VNIs, VRFs, Anycast Gateway and ECMP, with emphasis on scalable east-west connectivity and multi-tenant segmentation.
- Engineered advanced routing and connectivity architectures using BGP, MPLS, SD-WAN, IPsec, GRE, DMVPN, FlexVPN and WireGuard across distributed network environments.
- Designed multi-tenant ISP and enterprise architectures using VRF-based segmentation, centralized management, secure service isolation and high-availability principles.
- Engineered network observability and CPE management platforms using GenieACS/TR-069, SolarWinds, Grafana, Prometheus, Zabbix, LibreNMS and PRTG for centralized monitoring and lifecycle management.
- Deployed virtualized and containerized network services using VMware ESXi, Proxmox, Docker and Linux, integrating infrastructure monitoring, remote management and automation capabilities.

3%

July 2024 – March 2025

IT Infrastructure Engineer

Managed complete network infrastructure: routing, switching, firewall policy enforcement, and endpoint connectivity. Deployed NGFW policies on Palo Alto and Fortinet. Administered Windows Server (AD, DNS, DHCP, GPO) and VMware ESXi environments. Established centralized monitoring with Zabbix, Grafana, and Prometheus.

StormFiber

Feb 2024 – July 2024

Network Support Engineer

- Engineered L1–L3 troubleshooting for residential fiber (GPON/FTTH) customers. Resolved BGP peering, OSPF adjacency, and route propagation issues on access/aggregation layers. Configured VLANs, QinQ, and PPPoE on OLT/BNG platforms. Performed optical power budget analysis and fiber splicing verification.

Education

University of Engineering and Technology, Taxila

2020 - 2024

B.Sc. Electrical Engineering — Communication Engineering

PakTurk Maarif Int’l School & College, Islamabad

2018 - 2020

F.Sc Pre-Engineering

Certifications

Cisco Certified Network Associate (CCNA)

Linux Foundation — LFS114

Cisco — Network Support & Security

Cisco — Network Addressing & Basic Troubleshooting

Cisco — Networking Basics

Cisco — Linux Essentials